CANDIDATE NAME (CG) ________________________ **INDEX NUMBER** _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2008

BIOLOGY
Higher 2

Paper 3 (Applications) - 9747

Wednesday
27 August 2008

1 hour 30 min

Additional materials: Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
Write your answers in spaces provided on the question paper.

At the end of examination,
1. fasten all your work securely together;

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
Section B	
1	**/10**
2	**/10**
3	**/10**
4	**/15**
Section B	
5	**/20**
TOTAL	**/65**

Section A (45 m)

Answer all questions in the space provided

Question 1

Restriction enzymes cut DNA molecules only at specific sites with particular base sequences. The target sites for the enzyme EcoR1, Bam H1 and Hpa 1 are shown in Table 1.1 below. The lines drawn in each sequence show where the enzyme cuts the DNA molecule.

restriction enzyme	specific target base sequence of DNA
EcoRI	G A A T T C C T T A A G
BamHI	G G A T C C C C T A G G
HpaI	G T T A A C C A A T T G

Table 1.1

(a) State the type of ends created when cutting DNA with the restriction enzyme.

(i) EcoR1: .... [1]

(ii) Hpa 1:... [1]

(b) The target base sequences for all three restriction enzymes in Table 1.1 are palindromic.

(i) With reference to Table 1.1, explain what is meant by the term palindromic. [1]

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(ii) Suggest why it is preferable that restriction enzyme target sequences be palindromic. [1]

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A bacterial plasmid includes the base sequence shown in Figure 1.2 below. There are no target sequences for the three restriction enzymes in the rest of the plasmid.

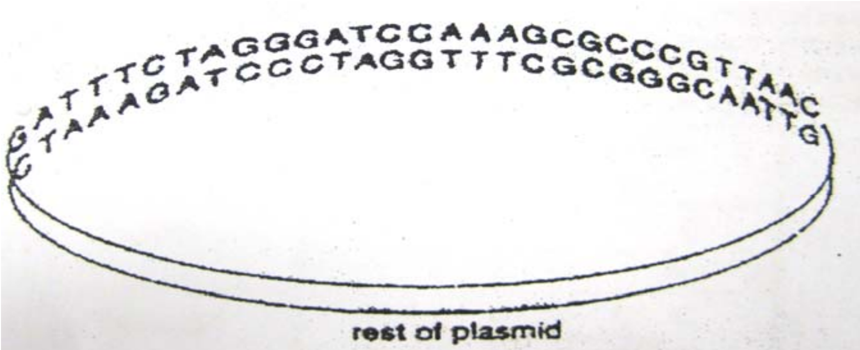

Figure 1.2

(c) State how many fragments of DNA will be present after the plasmid has been treated with:

(i) EcoR1: ....[1]

(ii) a mixture of EcoR1 and BamH1: ....[1]

A genetic engineer would like to use the above plasmid in genetic engineering.

(d)(i) Explain why he needs to use a plasmid for the process of genetic engineering.[1]

....

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(ii) Suggest which enzyme from Table 1.1 should he use and explain why this enzyme would be the scientist's choice. [3]

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[10 marks]

Question 2

Figure 2.1 shows the three stages involved in the polymerase chain reaction (PCR).

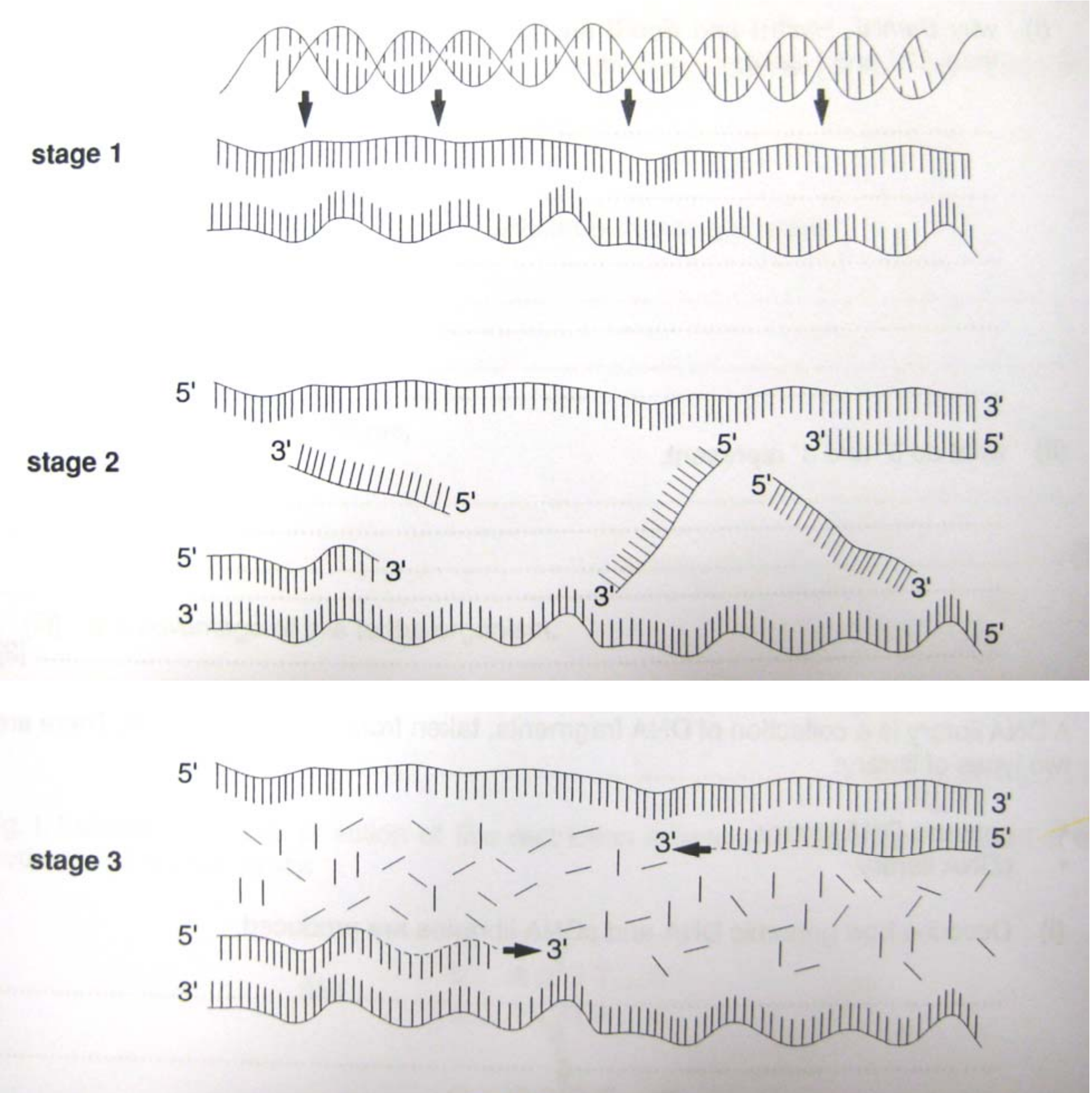

Fig. 2.1

With reference to Figure 2.1, explain what is happening at each of the three stages.

(a) stage 1 [2]

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stage 2 [2]

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stage 3 [2]

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(b) Why is it necessary to join primer molecules to the single stranded DNA before DNA polymerase is used? [1]

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(c) State 3 ways in which PCR differs from DNA replication in living cells. [3]

PCR	**DNA replication**

[10 marks]

Question 3

(a) The family with the autosomal dominant nail-patella syndrome required a second evaluation because individual 3 from generation II could not have a prenatal diagnosis using new restriction fragment length polymorphisms (RFLPs) based on her husband's genotype. **Fig. 3.1** illustrates a polymerase chain reaction (PCR) analysis that examines a variable number of tandem repeats (VNTRs) in a DNA region very near the nail-patella locus. Each repeat contains two nucleotides, and the number of repeats varies to produce at least six differently sized alleles when the region is amplified by PCR, using primers P1 and P2.

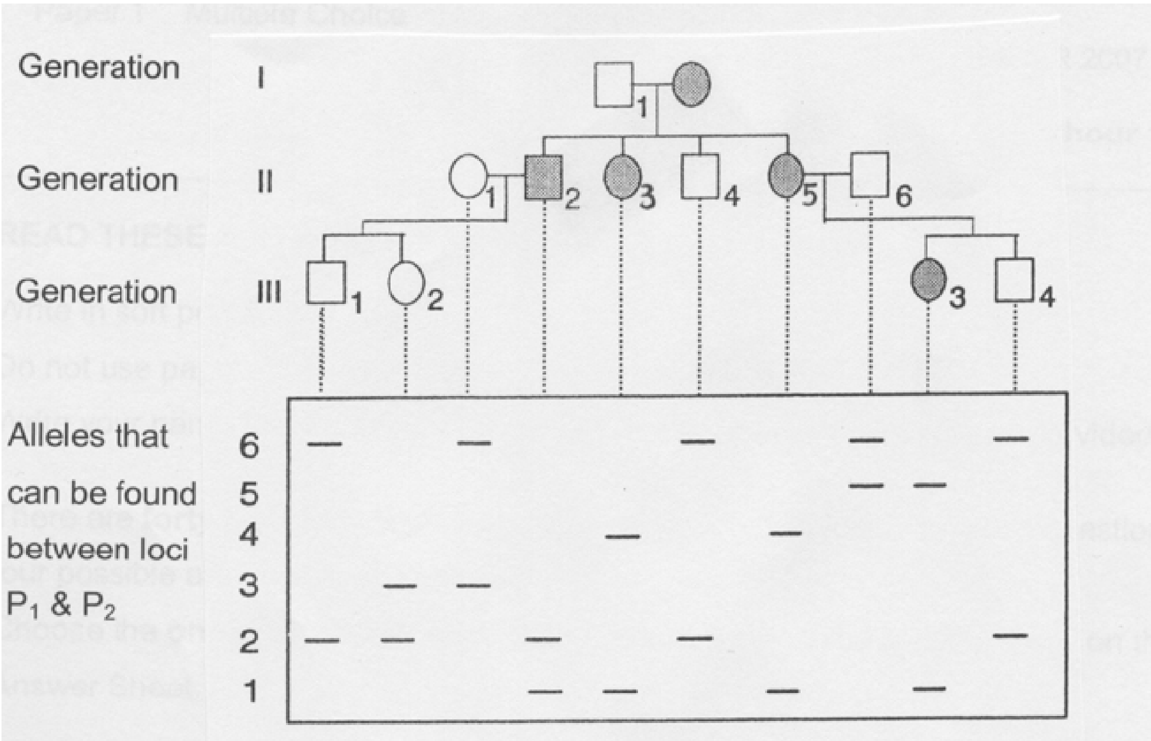

Fig. 1.1

(i) Explain the molecular basis for a restriction fragment length polymorphism.[2]

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(ii) Based on the family analysis shown in the figure, match the individual 3 from generation II with correct genotype. [1]

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(iii) Outline how an RFLP can be detected experimentally. [3]

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(b) Species A contain two different RFLPs which are linked.
RFLP 1 is found in 4500 bp and 5200 bp lengths and RFLP 2 can be 2100 bp or 3200 bp. A homozygote, 4500 bp and 2100 bp, is crossed to a homozygote, 5200 bp and 3200 bp. The F_1 offspring are then crossed to a homozygote containing the 4500 bp and 2100 bp fragments. The following results are obtained:

5200, 4500 and 2100	40 offspring
5200, 4500, 3200 and 2100	98 offspring
4500 and 2100	97 offspring
4500, 2100 and 3200	37 offspring

(i) State the genotypes of the recombinant offspring.[1]

1. ……………………………………..

2. ………………………………………

(ii) Calculate the map distance between the RFLPs.[1]

(iii) Explain how RFLP analysis facilitated the process of genomic mapping.[2]

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[10 marks]

Question 4

The following **Figure 4.1** illustrates how stems cells are cultivated for research.

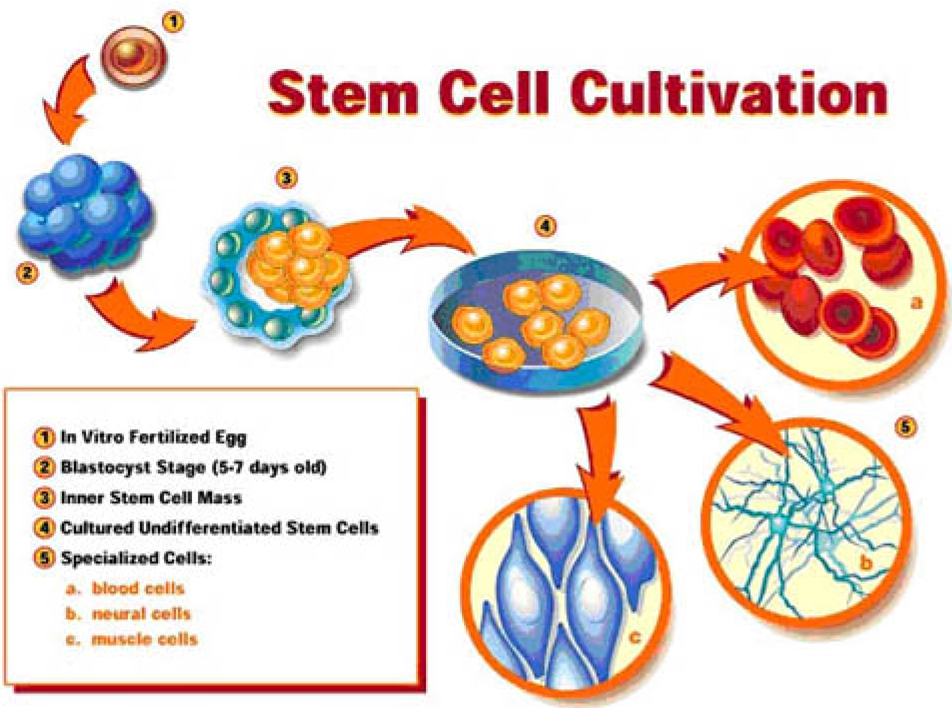

Figure 4.1

(a) Explain what makes a stem cell unique from a normal human adult cell. [3]

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(b) Name the chemical required to convert the undifferentiated stem cells in Step 4 of Figure 1.1 into the specialised cells in Step 5.[1]

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(c) The stem cells cultivated above are from the inner cell mass of the blastocyst. Are these the most useful stem cells available to science? Explain your answer. [3]

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(d) State and explain one similarity between a stem cell and a cancer cell.[2]

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(e) (i) Discuss the two ways in which SCID can come about in an individual.[4]

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[Turn over

(e) (ii) Suggest how stem cells could be used to enhance gene therapy for patients suffering from Severe Combined Immunodeficiency (SCID). [2]

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[Total: 15 marks]

Section B (20 marks)

Question 5

(a) Outline the procedure and discuss the underlying principle for the process of Southern blotting. [15]

(b) Compare the process of Western and Southern blotting. [5]